

Homework 5 - MAT 226

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Exercise 1.9

Q2) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

$$T(\vec{e}_1) = (1, 4) \quad T(\vec{e}_2) = (-2, 9) \quad T(\vec{e}_3) = (3, -8)$$

$$T = [T(\vec{e}_1) \quad T(\vec{e}_2) \quad T(\vec{e}_3)]$$

$$= \begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}$$

Q4) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

standard basis vectors $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$T(\vec{e}_1) = \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

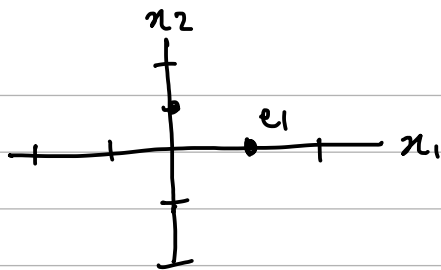
$$T(\vec{e}_2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A = [T(\vec{e}_1) \quad T(\vec{e}_2)] = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Q10) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

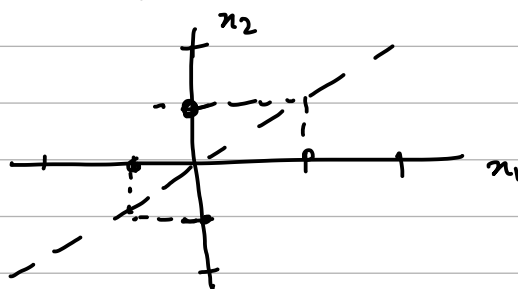
① reflect through horizontal x_1 -axis



$\vec{e}_1$ lies on x_1 so it remains unchanged $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\vec{e}_2$ is reflected in x_1 $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

(2) reflect through line $x_2 = x_1$



$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Q24) (a) False

The keyword here is 'onto'. For a transformation to map onto $\mathbb{R}^4$, the columns of the standard matrix must span $\mathbb{R}^4$ (matrix needs 4 pivots). 4×3 matrix has 3 columns so 3 max pivots.

The transformation maps $\mathbb{R}^3$ into $\mathbb{R}^4$ but not onto it.

(b) True

Directly from theorem, for any valid linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, there exists a unique $m \times n$ standard matrix such that $T(\vec{x}) = A\vec{x}$

for all $\vec{x}$.

(c) True
Standard matrix A is formed by $T(\vec{e}_1) T(\vec{e}_2) \dots T(\vec{e}_n)$ and assembling them as columns.

where $\vec{e}$ are the columns of $n \times n$ matrix.

(d) False

To be one-to-one, each output vector in $\mathbb{R}^m$ must be mapped by at most one input vector in $\mathbb{R}^n$.

(e) False

horizontal shear factor is at first row & second column for a 2×2 matrix.

26)

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 4R_1$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 17 & -20 \end{bmatrix}$$

One-to-One

Linear transformation is one-to-one if & only if its standard matrix has no free variables. Here we have 3 columns and 2 rows so 2 pivots at max.
 $A\vec{x} = 0$ has non trivial solutions.

Onto

A linear transformation would map $\mathbb{R}^3$ onto $\mathbb{R}^2$ if and only if the columns of its standard matrix span $\mathbb{R}^2$.

Matrix has pivot position in every row hence it spans $\mathbb{R}^2$.

The linear transformation is onto but not one-to-one.

32) T map $\mathbb{R}^n$ onto $\mathbb{R}^m$ if & only if A has m pivot columns.

Theorems:- $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A (standard matrix) span $\mathbb{R}^m$

Also, columns of $m \times n$ matrix span $\mathbb{R}^m$ if and only if A has a pivot position in every single row.

For the matrix to have pivot in every single row, it must have m pivots.

Since every pivot is in a unique column, having m pivot positions means having m pivot columns.

Exercise 2.1

$$Q10) \quad A = \begin{bmatrix} 3 & -6 \\ -1 & 2 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 1 \\ 3 & 4 \end{bmatrix} \quad C = \begin{bmatrix} -3 & -5 \\ 2 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 & -6 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \times -1 + -6 \times 3 & 3 \times 1 + -6 \times 4 \\ -1 \times -1 + 2 \times 3 & -1 \times 1 + 2 \times 4 \end{bmatrix} = \begin{bmatrix} -21 & -21 \\ 7 & 7 \end{bmatrix}$$

$$AC = \begin{bmatrix} 3 & -6 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -3 & -5 \\ 2 & 1 \end{bmatrix}$$

$$AC = \begin{bmatrix} 3x-3+2x-6 & 3x-5+1x-6 \\ -1x-3+2x2 & -1x-5+2x1 \end{bmatrix}$$

$$AC = \begin{bmatrix} -21 & -21 \\ 7 & 7 \end{bmatrix}$$

$$AB = AC = \begin{bmatrix} -21 & -21 \\ 7 & 7 \end{bmatrix}$$

$$\text{but } B \neq C$$

$$\begin{bmatrix} -1 & 1 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} -3 & -5 \\ 2 & 1 \end{bmatrix}$$

$$12) \quad A = \begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{let } B = [\vec{b}_1 \quad \vec{b}_2]$$

$$AB = A [\vec{b}_1 \quad \vec{b}_2] = [A\vec{b}_1 \quad A\vec{b}_2]$$

$$A\vec{b}_1 = 0$$

$$A\vec{b}_2 = 0$$

$$\begin{bmatrix} 3 & -6 & 0 \\ -2 & 4 & 0 \end{bmatrix}$$

$$R_1 \leftarrow R_1 / 3$$

$$\begin{bmatrix} 1 & -2 & 0 \\ -2 & 4 & 0 \end{bmatrix}$$

$$R_2 \leftarrow R_2 + 2R_1$$

$$\begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 2x_2$$

x_2 is a free variable

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

for $\vec{b}_1$, let $x_2 = 1$

$$1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

for $\vec{b}_2$, let $x_2 = 2$

$$2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \times 2 + 1 \times -6 & 3 \times 4 + 2 \times -6 \\ -2 \times 2 + 4 \times 1 & -2 \times 4 + 2 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

16) (a) True

first row of AB is formed by taking the first row in A & multiplying with B .

(b) False

$$AB = [A\vec{b}_1 \quad A\vec{b}_2 \quad A\vec{b}_3]$$

$$\text{and not } [A\vec{b}_1 + A\vec{b}_2 + A\vec{b}_3]$$

(c) True

from theorem we know that

$$(AB)^T = B^T \cdot A^T$$

if $B = A$ then

$$(AA)^T = A^T \cdot A^T$$

$$(A^2)^T = (A^T)^2$$

(d) False

$$(ABC)^T = C^T \cdot B^T \cdot A^T$$

the given expansion is incorrect as it swaps the position of A^T & B^T .

(e) True

$$(A+B)^T = A^T + B^T$$

Q22) if columns of B are linearly independent,

$B\vec{x} = \vec{0}$ has a non trivial solution.

to show that AB is linearly independent,
we must show that $(AB)\vec{x} = \vec{0}$

$$(AB)\vec{x} = \vec{0}$$

$$(AB)\vec{x} = A(B\vec{x})$$

$$B\vec{x} = \vec{0}$$

$$A(B\vec{x}) = A(\vec{0}) = \vec{0}$$

Any matrix multiplied by the zero vector produces $\vec{0}$.

Hence, we showed that $(AB)\vec{x} = \vec{0}$

Therefore, the columns of matrix AB are
linearly independent.